

Problem

A source is connected to three loads Z_1 , Z_2 , and Z_3 in parallel. Which of these is not true?

- (a) $P = P_1 + P_2 + P_3$
- (b) $Q = Q_1 + Q_2 + Q_3$
- (c) $S = S_1 + S_2 + S_3$
- (d) $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3$

Step-by-step solution

Step 1 of 5

Draw the circuit with a source connected to three loads Z_1 , Z_2 , and Z_3 as shown in the following figure:

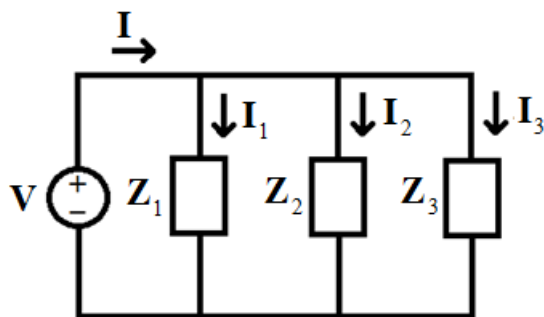


Figure 1

Step 2 of 5

Apply the Kirchhoff's current law to the circuit.

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3$$

The complex power provided by the source is,

$$\begin{aligned} \mathbf{S} &= \mathbf{V}\mathbf{I}^* \\ &= \mathbf{V}(\mathbf{I}_1^* + \mathbf{I}_2^* + \mathbf{I}_3^*) \\ &= \mathbf{V}\mathbf{I}_1^* + \mathbf{V}\mathbf{I}_2^* + \mathbf{V}\mathbf{I}_3^* \\ &= \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3 \end{aligned}$$

Here, $\mathbf{S}_1$, $\mathbf{S}_2$, and $\mathbf{S}_3$ are the individual complex powers provided to the circuit.

Therefore, option (d) is not correct.

Step 3 of 5

The complex power provided by the source is,

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3$$

Therefore,

$$P + jQ = (P_1 + jQ_1) + (P_2 + jQ_2) + (P_3 + jQ_3)$$

$$P + jQ = (P_1 + P_2 + P_3) + j(Q_1 + Q_2 + Q_3) \dots\dots (1)$$

Compare the real terms on both the sides.

$$P = P_1 + P_2 + P_3$$

Therefore, option (a) is not correct.

Step 4 of 5

Recall equation (1).

$$P + jQ = (P_1 + P_2 + P_3) + j(Q_1 + Q_2 + Q_3)$$

Compare the imaginary terms on both the sides.

$$Q = Q_1 + Q_2 + Q_3$$

Therefore, option (b) is not correct.

Step 5 of 5

The complex power provided by the source is,

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3$$

The apparent power provided by the source is,

$$\begin{aligned} S &= |\mathbf{S}| \\ &= \sqrt{P^2 + Q^2} \end{aligned}$$

The apparent power provided to load 1 is,

$$\begin{aligned} S_1 &= |\mathbf{S}_1| \\ &= \sqrt{P_1^2 + Q_1^2} \end{aligned}$$

The apparent power provided to load 2 is,

$$\begin{aligned} S_2 &= |\mathbf{S}_2| \\ &= \sqrt{P_2^2 + Q_2^2} \end{aligned}$$

The apparent power provided to load 3 is,

$$\begin{aligned} S_3 &= |\mathbf{S}_3| \\ &= \sqrt{P_3^2 + Q_3^2} \end{aligned}$$

Therefore,

$$S \neq S_1 + S_2 + S_3$$

Therefore, the correct option is ☐ c .